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Can digital technology break the financing dilemma of innovative SMEs? Based on the perspective of enterprise innovation

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ABSTRACT

Based on Signaling theory, this paper examines the mechanisms and effects of SMEs adopting digital technology to alleviate their own financing constraints from an innovation perspective. Empirical testing was conducted using data from enterprises in Zhongguancun, China, spanning the years 2014 to 2019. The study finds that the application of digital technology can effectively alleviate the financing constraints faced by enterprises. Additionally, the impact varies depending on the specific digital technology employed and the level of its adoption. The mechanism analysis reveals that enhancing innovation capability and integrating into innovation networks are crucial pathways through which digital technology alleviates financing constraints. Specifically, the application of digital technology improves firms' innovation capabilities, facilitates their integration into innovation networks, attracts external investment by effectively signaling high-quality attributes to the market, and thereby mitigates financing constraints. Further analysis of heterogeneity indicates that the alleviating effect of digital technology on financing constraints is more pronounced when entrepreneurs' cognitive abilities are stronger, when industry competition is intense, and when enterprise scale is larger. This paper provides an empirical foundation for addressing SMEs' financing dilemmas in the digital era and supports high-quality development of enterprises.

1. Introduction

Small and medium-sized enterprises (SMEs) play a crucial role in enhancing employment opportunities and driving economic growth, acting as a significant market force and an essential pillar for economic development (Lim et al., 2021). Notably, SMEs account for 90 % of all businesses and contribute to 60–70 % of employment across various economies (Soni et al., 2022). Despite their importance, SMEs often face persistent financing challenges that represent a substantial obstacle to their sustainable development. By the end of 2017, the loan balance allocated to SMEs in China constituted merely 24.67 % of the total bank loan portfolio. Approximately 23 million small and medium-sized enterprises encountered general financing constraints, with a financing gap rate reaching 43 %. Among these entities, 33 % of medium-sized enterprises, 38.8 % of small enterprises, and 40.7 % of micro-enterprises were unable to fulfill their financing requirements. Therefore, addressing the financial challenges faced by SMEs is imperative for fostering enterprise growth and enhancing overall economic vitality (Lim et al., 2021; Soni et al., 2022).

The existing literature provides valuable insights into the impact of digital technologies on corporate financial practices and outcomes. Some studies suggest that enterprise digital transformation can effectively alleviate financing constraints by reducing information asymmetry, addressing principal-agent issues, lowering transaction costs, enhancing the quality of information disclosure, and improving external communication environments (Xue et al., 2022; He et al., 2023; Guo et al., 2023). However, other literature argues that digital technology may also have a negative impact on alleviating financial constraints (Triplett, 1999; Porter and Heppelmann, 2014; Sjödin et al., 2020; Cepa, 2021). Many enterprises in the early stages of digital transformation often lack a comprehensive understanding of the digital economy, which hampers their ability to fully grasp the true effects of this transformation

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on enterprise development (Triplett, 1999). Furthermore, digital transformation typically requires substantial long-term investments—such as ongoing implementation and maintenance costs—that are difficult for enterprises to offset in the short term (Porter and Heppelmann, 2014; Sjödin et al., 2020). These increased short-term expenditures can constrain developmental funds within an enterprise potentially undermining short-term profitability and adversely affecting access to financing.

The aforementioned literature has provided valuable insights into the mitigating effects of digital technology on financial constraints. However, a significant portion of studies focusing on the Chinese context predominantly examines listed companies, which typically garner more attention and enjoy comparatively easier access to financing than unlisted SMEs. Furthermore, these investigations may have overlooked the critical role that innovation plays in the financing process.

The lack of standardized financial reporting systems and external stakeholder oversight presents significant challenges for investors in accurately assessing the true quality of SMEs (Shane and Stuart, 2002). This situation creates substantial information barriers between SMEs and external parties. The advent of the digital economy has enhanced information processing efficiency and transformed enterprise management practices, thereby offering new avenues to mitigate these information barriers. However, these developments are fair: while a reduction in information asymmetry can facilitate access to financing opportunities for high-quality enterprises, it may simultaneously intensify financing constraints for lower-quality firms. Given that the quality of start-ups is often not directly observable, external investors must depend on alternative sources of information, particularly the discernible characteristics of new businesses (Stuart et al., 1999). Consequently, SMEs lacking credit guarantees must not only dismantle information barriers with less informed parties but also effectively communicate high-quality signals to demonstrate their robust development potential. This approach will enhance their chances and opportunities for securing investment.

Innovation capabilities and the integration of enterprises into innovation networks can serve as high-quality signals of a company's technological prowess and potential, thereby enhancing external perceptions and expectations of the enterprise. Patents, as a direct reflection of an enterprise's innovation ability and potential intangible assets, possess observable, non-mimetic, and technically exclusive characteristics, allowing them to be transmitted as high-quality signals to the outside world. Numerous studies have demonstrated that patents can serve as powerful signals to attract external investment (Mann and Sager, 2007; Haeussler et al., 2012; Hsu and Ziedonis, 2013; Hoenen et al., 2014). Moreover, the innovation networks in which enterprises participate are also important signals for external investors to assess the quality of the enterprise (Stuart, 2000). Although financing difficulties and high costs are common issues for SMEs, firms that are more innovative and competitive have easier access to trade credit (Klapper and Randall, 2011). In recent years, the vigorous development of emerging industries centered around digital technology has led to the emergence of a series of rapidly growing unicorn companies. These companies have garnered enthusiastic pursuit from numerous venture capital investments due to their innovative technologies and business models. Therefore, drawing on signaling theory from an innovation perspective, this paper aims to investigate the impact of digital technology application on the financing constraints of SMEs.

Considering the above, this paper utilizes a dataset from enterprises in the Zhongguancun Haidian Science and Technology Park (ZHSP) and employs a staggered Difference-in-Differences (DID) model to empirically test the impact of digital technology application on corporate financing constraints. The study finds that the application of digital technology can effectively alleviate corporate financing constraints, with the effects varying according to the type and level of digital technology adopted. Innovation capabilities and integration into innovation networks are identified as significant pathways through which digital technology mitigates financing constraints. The research indicates that the application of digital technology can enhance a company's innovation capabilities, facilitate its integration into innovation networks, and effectively transmit high-quality signals to external stakeholders, thereby attracting external investment and alleviating corporate financing constraints. Furthermore, differences in enterprise characteristics and development environments lead to heterogeneous effects of digital technology application on corporate financing constraints. The study finds that the alleviating effect of digital technology application on financing constraints is stronger when industry competition is more intense, the entrepreneur's cognitive ability is stronger, and the enterprise size is larger. Based on these research findings and in conjunction with the actual situation, we propose several recommendations to assist SMEs in alleviating financing constraints.

The remainder of our study proceeds as follows: Section 2 elaborates on the theoretical mechanism and proposes the research hypothesis from the perspective of signaling theory. Section 3 introduces the data and models utilized in this empirical demonstration. Section 4 presents the main empirical results, including heterogeneity analysis and robustness tests. Section 5 discusses the empirical testing of innovation as an intermediary mechanism. Finally, Section 6 summarizes the conclusions of this paper.

2. Theoretical mechanisms and research hypotheses

2.1. The impact of digital technology on innovative SMEs financing

Information asymmetry is a critical factor constraining SMEs' financing and serves as a key obstacle to the transmission of high-quality signals from within the enterprise to the outside world. The application of digital technologies, however, offers potential for SMEs to break through information barriers and alleviate information asymmetry both internally and externally. Internally, the adoption of digital technologies drives upgrades and restructuring in organizational structures, production models, product design, and employment patterns (Nambisan, 2017). As the degree of digitization deepens within SMEs, data elements retained from business operations become a driving force for enhancing digital governance capabilities. This enhances SMEs' ability to process large-scale, unstructured, and non-standardized data, significantly improving information processing levels, circulation efficiency, and transparency. This means that digital technologies can improve how SMEs handle and output effective information, enabling them to "push" information to market entities more effectively (Wu et al., 2021). Additionally, SMEs can more easily publish standardized information for external identification and analysis, bringing advantages such as increased market attention and improved market expectations (Wu et al., 2024). Moreover, automated and standardized data processing not only helps enhance the quality of basic financial data but also reduces information asymmetry between the company and shareholders or creditors, strengthening internal control quality and limiting executives' discretion over accounting data or policies, thereby improving the quality of information disclosure.

Externally, as digital technologies penetrate and integrate into various aspects of SMEs' production, operation, management, and transactions, offline business activities can be mirrored online, leaving many "digital footprints" related to actual business operations (Zhang et al., 2021). These "digital footprints" provide financial institutions with new perspectives and tools for risk assessment, allowing them to consider multidimensional data sources such as online inventory orders and transaction records, rather than being limited to traditional financial data. The integration of digital technology with credit services broadens channels for information collection and enhances transparency for credit institutions (Liu et al., 2022). Furthermore, research shows that when SMEs start applying digital technologies, they often receive more attention from analysts and may become key tracking targets (Fang et al., 2023). This external attention can further help

mitigate information asymmetry between the inside and outside of SMEs, creating more favorable conditions for their financing activities. In summary, the application of digital technologies can exert efforts both internally and externally within SMEs, establishing effective digital information transmission channels. This allows high-quality internal information to be effectively conveyed outward, reducing the level of information asymmetry between SMEs and the outside world, and alleviating financing constraints.

From the perspective of internal quality characteristics within an enterprise, digital transformation offers several benefits. Firstly, it can reduce costs associated with production, distribution, transactions, and operations (Benner and Waldfogel, 2023; Yang et al., 2021). This reduction in costs enhances operational efficiency and transforms uncertainty into manageable opportunities. Additionally, it assists firms in identifying additional business prospects (Benner and Waldfogel, 2023), ultimately leading to an improvement in overall performance (Liu et al., 2022; Zhai et al., 2022). Furthermore, the digital economy alters business dynamics, institutional quality, and organizational structures (Yuan et al., 2021). It decreases search, replication, transportation, traceability, and authentication costs for enterprises while significantly enhancing production flexibility and risk management (Goldfarb and Tucker, 2019). Digital transformation within enterprises can mitigate operational leverage and risks by reducing business operation costs as well as enhancing market share and competitiveness (Liu et al., 2022). Moreover, the capability to undergo digital transformation has become a crucial pathway for gaining competitive advantages over traditional enterprises (Llopis-Albert et al., 2021; Ferreira et al., 2019; Annarelli et al., 2021). The application of digital technologies can accelerate internal quality, efficiency, and dynamism, strengthening external connections with markets, users, and other firms (Chavez et al., 2017), enabling faster responses to market demands, thereby laying the foundation for enhanced competitiveness in complex market competitions.

In summary, on the one hand, the adoption of digital technology by enterprises can break down the existing information barriers between the firm and external stakeholders, open up information channels, and effectively reduce the level of information asymmetry, thereby alleviating financing constraints. On the other hand, digital transformation within enterprises can promote development by enhancing profitability, lowering risk levels, and boosting competitiveness, thus increasing the willingness of investors to invest. Accordingly, we propose the following hypothesis.

Hypothesis 1. The application of digital technology by companies can significantly ease financing constraints.

2.2. Mechanisms of innovation in influencing corporate financing constraints

Digital technology can effectively reduce the level of information asymmetry between enterprises and external stakeholders (Xue et al., 2022; He et al., 2023; Guo et al., 2023). However, merely reducing information asymmetry may not be sufficient to secure funding support. Enterprises also need to convey high-quality signals to the outside world, demonstrating their high quality and strong development potential, in order to enhance external perceptions and expectations, and increase the likelihood of obtaining financing or venture capital. Based on signaling theory, entities with information advantages can transmit high-quality signals to those with information disadvantages to alleviate the information asymmetry between them. Signals can be categorized as strong or weak and may be subject to interference from noise. Stronger signals are more effective in conveying information and can better improve external expectations of the enterprise (Spence, 1973; Lampel and Shamsie, 2000; Spence, 2002).

SMEs, due to their lack of sufficient physical collateral, often struggle to obtain direct financing from credit markets. Innovation capabilities and integration into innovation networks are important sources of competitive advantage for SMEs and are key factors influencing their future performance. These capabilities can serve as high-quality signals of a company's technological quality and potential, thereby improving external perceptions and expectations of the enterprise.

Patents serve as a tangible manifestation of a corporation's innovative prowess and its intangible assets, marked by their observability, non-imitability, and technological exclusivity. These characteristics transmit high-quality signals to stakeholders. Innovation not only reduces production costs but also enhances organizational performance, thereby serving as a critical source of competitive advantage for firms (Cui et al., 2015). Companies endowed with such competitive advantages can sustain profitability amidst uncertain external market conditions, thereby reflecting the potential financial strength and solvency of SMEs, which in turn bolsters investor confidence. Extensive literature supports the notion that patents act as potent signals to attract external investment (Mann and Sager, 2007; Haeussler et al., 2012; Hsu and Ziedonis, 2013; Hoenen et al., 2014).

Moreover, the integration of companies into innovation networks constitutes another significant signal for external investors to gauge corporate quality. Through endorsements from universities and research institutions, companies can emit positive signals regarding their technological caliber, thereby enhancing their investment attractiveness and increasing the likelihood of securing venture capital (Hoenig and Henkel, 2015). Additionally, to safeguard their reputations, these external institutions tend to select their partnerships judiciously, making the selection itself a positive signal to the broader public (Stuart, 2000). Consequently, the incorporation of firms into innovation networks serves as a pivotal technological and quality signal to draw external investors.

Entering the digital age, the signaling effect of innovation on corporate financing has become particularly pronounced. Digital technologies have fundamentally transformed the co-creation of value among innovation entities, their organizational structures, and methods of collaboration, while continuously enhancing the openness and synergy of innovation, reshaping the innovation ecosystem.

From the perspective of enterprise technological innovation processes, digitization has altered how companies acquire, store, and utilize knowledge (Joshi et al., 2010). The application of digital technologies aids enterprises in collecting information on hidden customer needs and behavioral preferences (Mrázová and Neary, 2017), making the generation of ideas more timely. With the development and application of digital technologies, data on networks have become machine-readable, rendering relevant information interrelated (Yoo, 2010), which facilitates easier knowledge searches for enterprises. Digital technologies can assist in creating and testing new knowledge combinations (Joshi et al., 2010), promoting knowledge recombination and ultimately achieving technological innovation. Enhanced corporate technological innovation capabilities can send positive signals to the market. Patents resulting from these innovative activities serve as visible disclosures, reducing information asymmetry between the company and external stakeholders, thereby alleviating financing constraints.

Additionally, the application of digital technologies plays a crucial role at various stages of the enterprise product innovation process. In the front-end stage of product innovation, enterprises can leverage Big Data and Cloud Computing technologies to conduct precise analyses of market demand, technology trends, and industry dynamics, thereby more effectively identifying potential innovation opportunities (Rialti et al., 2018; Benner and Waldfogel, 2023). During the product development phase, these advanced digital methods significantly enhance R&D efficiency, reduce trial-and-error costs, and shorten product development cycles. The integration of the Internet of Things (IoT) and Intelligent Hardware facilitates interoperability between different types of information and transforms processed information into knowledge within the Internet of Everything (Vermesan and Friess, 2014), aiding in optimizing design, identifying defects, and accelerating the speed of bringing products to market (Blichfeldt and Faullant, 2021). In the

commercialization stage, digital technologies improve the company's market response speed and customer demand responsiveness, reducing the costs associated with market expansion and customer acquisition (Chavez et al., 2017). Especially during the promotion of high-quality innovations, digital transformation can effectively spread innovation costs and increase marginal returns, providing sustained competitive advantages for enterprises (Llopis-Albert et al., 2021; Ferreira et al., 2019; Annarelli et al., 2021). This enhancement in product innovation capability also signals future profitability and business growth potential to the outside world, attracting attention from external investors.

In terms of integrating into innovation networks, digital technologies overcome geographical and resource barriers, expanding the scope and depth of innovation activities and significantly strengthening connections between innovation entities. IoT and Intelligent Hardware allow innovation activities to occur beyond centralized locations, while leveraging Big Data and Cloud Computing technologies' distributed computing and storage models, enterprises can share information with other innovation entities (such as research institutions and innovation alliances), promoting cross-regional and cross-departmental collaboration. This benefits the regional reorganization of innovation resources, thereby enhancing inter-enterprise collaborative innovation levels (Li et al., 2022). Digital technologies can greatly enhance knowledge exchange and sharing, promoting innovation cooperation between enterprises (Peng and Tao, 2022), forming tightly coordinated networks (Cui et al., 2015). Enterprises deeply embedded in innovation networks can use this as a signal of technical quality, improving investment recognition and increasing opportunities for venture capital (Hoenig and Henkel, 2015).

In summary, based on signaling theory, digital technologies promote corporate financing from both internal and external perspectives. Internally, digital technologies foster corporate innovation and integration into innovation networks, enhancing corporate capabilities and competitive advantages, enabling them to send high-quality signals externally. Externally, digital technologies strengthen communication channels between enterprises and external stakeholders, making the company's high-quality signals and growth potential more evident, thus increasing the likelihood of obtaining financing and alleviating financing constraints.

Hypothesis 2. The adoption of digital technology can bolster a company's innovation capabilities and facilitate its integration into innovation networks, effectively signaling high quality to external parties, attracting external investment, and alleviating financing constraints.

3. Method

3.1. Data

The data utilized in this study are derived from a comprehensive dataset of enterprises located in ZHSP, Beijing, spanning the period from 2014 to 2019. This dataset encompasses a wide range of information, including basic details, operational and production metrics, financial statuses, R&D innovations, and human resources data for enterprises. Notable features of this dataset include its large sample size, extensive coverage of enterprise production and operations, extended temporal coverage. Its high reliability and representativeness make it a valuable and authoritative resource for the analysis of corporate economics.

The data is also advantageous in this study in the following three aspects: Firstly, compared to most existing literature on empirical research based on Chinese data, which predominantly feature samples from A-share listed enterprises—often larger and financially stronger, thus receiving preferential treatment from financial institutions—the dataset includes all enterprises within ZHSP, encompassing a significant number of private enterprises and SMEs. This mitigates potential endogeneity issues related to sample self-selection, thereby reducing estimation errors typically associated with selective sampling. Secondly, ZHSP, as a forefront player in digital economy development, exhibits a broader penetration of digital technology and a deeper level of enterprise digital transformation. This extensive digital adoption makes the dataset highly representative for studying the impact of digital technology on corporate financing. Thirdly, as a national demonstration zone for independent innovation, ZHSP benefits from robust policy support for enterprise innovation. Consequently, innovation activities in this region are more dynamic, and the integration of innovation into daily operations is more pronounced. This characteristic enhances the representativeness of the dataset for examining how digital technology influences the financing constraints faced by SMEs from an innovation perspective.

In accordance with the research requirements, the screening and processing of the necessary data for this study were carried out as follows: (1) Excluding enterprises in non-operational statuses such as those that have ceased operations, are in the preparatory phase, have been cancelled, or have filed for bankruptcy; (2) Removing enterprises with missing, zero, or negative values for key indicators such as total assets, primary business revenue, total number of employees, and firm age; (3) Winsorizing the main continuous variables at the 1st and 99th percentiles to address outliers. Following this process, a total of 49,600 firm-year sample observations from the unbalanced panel data for the period 2014–2019 were obtained. This included 13,139 sample observations with digital technology applied and 36,461 sample observations without digital technology applied.

3.2. Model and measures

Drawing upon existing studies, this paper proposes to investigate the impact of firms' adoption of digital technology on financing constraints using a staggered DID model.

$$FC_{it} = \beta_0 + \beta_1 Digital_{it} + \gamma X_{it} + \lambda_i + \theta_t + \varepsilon_{it} \quad (1)$$

where FC_{it} denotes the enterprise financing constraint. $Digital_{it}$ is the interaction term between whether the enterprise applies digital technology and before and after applying digital technology.

Where FC_{it} denotes the financing constraint of enterprise i at time t. The interaction term $Digital_{it}$ represents the interaction between whether the enterprise applies digital technology and the timing of its adoption. β_1 is the primary parameter of interest in this study, indicating the actual impact of an enterprise's application of digital technology on its financing constraints. If $\beta_1 < 0$, it suggests that the application of digital technology alleviates financing constraints; conversely, it indicates an intensification of financing constraints. X_{it} denotes a series of control variables, including enterprise revenue size, return on equity, asset-liability ratio, capital intensity, enterprise profitability, the proportion of R&D staff, and the nature of enterprise ownership. λ_i represents the industry fixed effect, θ_t denotes the time fixed effect, and ε_{it} represents the random disturbance term.

$$SA = -0.737size + 0.043size^2 - 0.040age \quad (2)$$

The dependent variable in this study is financing constraint (FC_{it}). While established studies commonly use indicators such as the WW index and KZ index to measure financing constraints, these measures are susceptible to endogenous variables such as cash flow and leverage. Therefore, we choose to utilize the SA index to measure corporate financing constraints, as demonstrated in model (2). The absolute value of the SA index is then employed as a measure of the financing constraint of the enterprise; a larger absolute value of the SA index indicates a greater level of financing constraint.

The independent variable in this study is the interaction term ($Digital_{it}$) between whether the firm applies digital technology and the timing of its application (before and after). The treatment and control groups are differentiated based on whether the enterprises apply digital technology: firms that adopt digital technology are coded as 1, while those

that do not are coded as 0. For the treatment group, the timing of digital technology adoption is further coded as 0 for the period before adoption and 1 for the period after adoption. According to the Statistical Reporting System of the Zhongguancun Independent Innovation Demonstration Zone, an enterprise is considered to be applying digital technology (coded as 1) if it utilizes any one of the following: Big Data, Cloud Computing, IoT, or Intelligent Hardware; otherwise, it is coded as 0.

In addition, based on theories related to corporate finance and the practices found in existing literature, this paper also incorporates the following control variables. Corporate revenue size, operating profit margin, and return on net assets are closely associated with financing constraints. Larger revenue size and higher profit levels indicate strong operating performance and profitability, which facilitate obtaining funds through internal financing (Zhang, 2020). Revenue size (*Income*) is expressed as the natural logarithm of the firm's total revenue. Operating profit margin (*Profit*) is represented as the ratio of operating profit to total corporate assets, and return on net assets (*ROE*) is expressed as the ratio of net profit to shareholders' equity. Furthermore, this paper controls for the proportion of R&D staff (*R&D_Staff*) and the nature of enterprise ownership (*SOE*). The proportion of R&D staff is measured by the ratio of the total number of R&D personnel to the total number of employees in the firm. Enterprise ownership nature is captured by a dummy variable: 1 if the enterprise is state-owned, and 0 otherwise. Given that capital intensity and gearing can significantly impact corporate financing (Bougheas et al., 2006; Zhang, 2020), this paper uses the natural logarithm of the ratio of fixed assets to the number of employees to measure capital intensity (*Cap*) and the ratio of total liabilities to total assets to express the gearing ratio (*Lev*).

4. Results

4.1. Descriptive statistics

A preliminary analysis was initially conducted using descriptive statistics to evaluate the impact of digital technology on the financing constraints of enterprises. Table 1 presents the descriptive statistics, with Panel A showing the descriptive statistics for each variable for companies applying digital technology, and Panel B showing the descriptive statistics for companies not applying digital technology. It is evident from the data that the mean, median, 5th and 95th percentile of the financing constraint variables for enterprises applying digital technologies are smaller than those for enterprises without digital technologies. This suggests that the financing constraints of enterprises adopting digital technologies may be relatively lower.

Table 1
Descriptive statistics of primary variables.

Panel A Descriptive statistics for companies applying digital technology						
Variables	*N*	*Mean*	*Std Dev*	*P5*	*Median*	*P95*
FC	13,139	3.2122	0.4375	2.3042	3.2913	3.7179
Income	13,139	9.9793	2.2546	5.7930	9.6912	13.5216
Cap	13,139	2.8516	1.6086	0.5341	2.6288	5.8693
Profit	13,139	−0.1172	0.4569	−0.9364	0.0025	0.2495
Roe	13,139	0.0071	0.9927	−1.1843	0.0342	1.0177
Lev	13,139	0.5948	0.7411	0.0352	0.4156	1.6973
R&D_Staff	13,139	0.4600	0.3034	0.0000	0.4622	1.0000
SOE	13,139	0.0865	0.2811	0.0000	0.0000	1.0000

Panel B Descriptive statistics for companies not applying digital technology						
Variables	*N*	*Mean*	*Std Dev*	*P5*	*Median*	*P95*
FC	36,461	3.3171	0.4091	2.5403	3.3826	3.8064
Income	36,461	9.3458	2.2407	5.3753	9.4348	12.9247
Cap	36,461	3.0247	1.7154	0.5108	2.8106	6.1932
Profit	36,461	−0.0636	0.3714	−0.5772	0.0030	0.2382
Roe	36,461	0.0327	0.8375	−0.7544	0.0232	0.7801
Lev	36,461	0.5949	0.6881	0.0342	0.4639	1.4917
R&D_Staff	36,461	0.3166	0.3193	0.0000	0.2632	0.9453
SOE	36,461	0.1012	0.3016	0.0000	0.0000	1.0000

Furthermore, this paper categorizes the samples into distinct control and experimental groups. To analyze the statistical significance of the mean financing constraints between these groups, a *t*-test is conducted. Initially, the number of invention patents is used to measure the innovation ability of enterprises, and the sample is divided into two parts based on the number of invention patents. Enterprises with a number of invention patent applications in the current year greater than the average value of the sample are categorized as having strong innovation ability; those below the average are categorized as having weak innovation ability.

Among enterprises with strong innovation ability, those that have applied digital technology form the experimental group, while those that have not applied digital technology form the control group. A *t*-test is then performed on the mean financing constraints of these two groups. The test results are presented in Group A of Table 2. These results show that, despite both groups having strong innovation ability, there was a significantly smaller mean financing constraint for enterprises in experimental group compared to those in control group. This finding preliminarily indicates that application of digital technology can help alleviate financing constraints for enterprises.

In addition, we altered the composition of the control group. Specifically, enterprises with weak digital technology application and innovation ability were used as the new control group to conduct a *t*-test. The test results are presented in Group B of Table 2. It was found that, even after changing the control group, the mean financing constraint of the experimental group remains significantly smaller. This suggests that businesses undergoing digital transformation, particularly those with stronger innovation capabilities, are likely to experience reduced financing constraints and potentially receive more funding. The above results preliminarily confirm our theoretical hypothesis.

4.2. Analysis of baseline regression results

Table 3 demonstrates the results of the benchmark regression analysis examining the impact of firms' application of digital technology on their financing constraints. In column (1), both industry fixed effects and time fixed effects are controlled. The results show that the regression coefficient for the application of digital technology by enterprises is −0.0156, which is significant at the 1 % level. Column (2) adds the previously mentioned control variables, including firm size, operating profit margin, return on net assets, proportion of R&D staff, nature of enterprise ownership, capital intensity, and gearing ratio, while still controlling for industry and time fixed effects. It can be seen that, although the coefficient in column (2) is different from that in column (1), it remains significant at the 5 % level. This indicates that the application of digital technology can significantly alleviate financing constraints for enterprises, suggesting that digital technology adoption inhibits financing constraints. Thus, Hypothesis 1 of this paper is

Table 2
T-statistic test for the dependent variable.

Panel A Without digital technology but with strong innovation ability are the control group				
Variables	Control Group	Experimental Group	Difference	T-values
FC	3.0898	2.9589	0.1309***	10.2084
N	5070	2778	–	–

Panel B With digital technology but with weak innovation ability are the control group				
Variables	Control Group	Experimental Group	Difference	T-values
FC	3.2802	2.9589	0. 3213***	28.9968
N	10,361	2778	–	–

Table 3
Regression estimation result.

	(1)	(2)
	FC	*FC*
Digital	−0.1064***	−0.0706***
	(0.0045)	(0.0040)
Income		−0.0846***
		(0.0012)
Cap		−0.0253***
		(0.0011)
Profit		0.1427***
		(0.0056)
Roe		−0.0032
		(0.0020)
Lev		−0.0134***
		(0.0031)
R&D_Staff		−0.0340***
		(0.0051)
SOE		−0.0623***
		(0.0082)
Cons	3.3175***	4.2185***
	(0.0021)	(0.0117)
Industry Fixed Effect	YES	YES
Time fixed effect	YES	YES
Within R^2	0.0118	0.2342
N	49,600	49,600

Note: ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$; Robust standard error are in parentheses, and the following tables are same.

supported.

Big Data, Cloud Computing, IoT, and Intelligent Hardware, when applied by enterprises, exhibit varying degrees and scopes of heterogeneity effects. Given the similarities between the IoT and Intelligent Hardware (Stojkoska and Trivodaliev, 2017; Gubbi et al., 2013), as well as the commonalities between Cloud Computing and Big Data in terms of data management and analysis (Botta et al., 2016), this paper categorizes these technologies into two groups. Specifically, IoT and Intelligent Hardware are grouped together due to their focus on real-world applications, while Cloud Computing and Big Data are grouped together for their emphasis on virtual resource management and data utilization.

To examine the heterogeneity effects, this paper constructs two binary variables representing the application of digital technology. If an enterprise uses either IoT or Intelligent Hardware, it is coded as 1; otherwise, it is coded as 0. Similarly, if an enterprise uses either Big Data or Cloud Computing, it is coded as 1; otherwise, it is coded as 0. These binary variables replace the original explanatory variables in the regression to test the differential impacts of these digital technologies on the alleviation of financing constraints.

Empirical results in columns (1) and (2) of Table 4 show that regardless of the type of digital technology adopted, it contributes to alleviating enterprises' financing constraints. However, the application of IoT and Intelligent Hardware is less effective in alleviating financing constraints compared to the application of Big Data and Cloud Computing. This may be because Cloud Computing and Big Data are more deeply integrated with core business processes, thus enhancing digital governance and promoting both enterprise development and access to financing.

Further analysis examines the impact of the extent of digital technology application on financing constraints. Two sets of binary variables are constructed: one where enterprises are coded as 1 if they apply any two of the four technologies, and another where enterprises are coded as 1 if they apply all four technologies simultaneously. Regression results in columns (3) and (4) of Table 4 demonstrate that enterprises that integrate all four digital technologies experience a more significant reduction in financing constraints. This indicates that deeper digital transformation leads to a stronger inhibitory effect on financing constraints, further confirming the significant role of digital technology in alleviating corporate financing constraints.

Table 4
Empirical Results on the Degree of Digitization on Corporate Financing Constraints.

	(1)	(2)	(3)	(4)
	FC	*FC*	*FC*	*FC*
Digital	−0.0523***	−0.0806***	−0.0866***	−0.1146***
	(0.0050)	(0.0044)	(0.0059)	(0.0163)
Income	−0.0858***	−0.0842***	−0.0849***	−0.0859***
	(0.0012)	(0.0012)	(0.0012)	(0.0012)
Cap	−0.0247***	−0.0253***	−0.0249***	−0.0246***
	(0.0012)	(0.0011)	(0.0011)	(0.0012)
Profit	0.1467***	0.1428***	0.1449***	0.1474***
	(0.0056)	(0.0056)	(0.0056)	(0.0056)
Roe	−0.0030	−0.0032	−0.0029	−0.0029
	(0.0020)	(0.0020)	(0.0020)	(0.0020)
Lev	−0.0125***	−0.0130***	−0.0128***	−0.0121***
	(0.0031)	(0.0031)	(0.0031)	(0.0031)
R&D_Staff	−0.0414***	−0.0346***	−0.0392***	−0.0453***
	(0.0051)	(0.0051)	(0.0051)	(0.0051)
SOE	−0.0605***	−0.0616***	−0.0620***	−0.0607***
	(0.0082)	(0.0082)	(0.0082)	(0.0082)
Cons	4.2188***	4.2128***	4.2121***	4.2150***
	(0.0118)	(0.0117)	(0.0118)	(0.0118)
Industry Fixed Effect	YES	YES	YES	YES
Time fixed effect	YES	YES	YES	YES
Within R^2	0.2309	0.2346	0.2330	0.2300
N	49,600	49,600	49,600	49,600

Furthermore, this paper further examines the impact of the application degree of digital technology on its own financing constraints. In this paper, the virtual variables of two digital technologies and four digital technologies are constructed respectively. If the enterprise applies any two of the above four technologies, it is marked as 1, otherwise it is 0. The variable construction method of applying the four digital technologies simultaneously is similar. Then, the above variables were replaced by the original explanatory variables for regression, and the empirical results were shown in columns (3) and (4) of Table 4. It can be seen from the results that if enterprises use four digital technologies at the same time, that is, deeply carry out digital transformation, they have a greater degree of mitigation of their own financing constraints. It shows that with the deepening of the use of digital technology, that is, the higher the level of digitalization of enterprises, the stronger the inhibitory effect of digital technology on alleviating corporate financing constraints, which further indicates that digital technology has a significant inhibitory effect on corporate financing constraints.

4.3. Heterogeneity test based on firm characteristics

4.3.1. Entrepreneur cognitive ability

According to the Upper Echelons theory (Hambrick and Mason, 1984), managers often encounter challenges in fully understanding their enterprise's circumstances due to the complexity of internal and external environments. As a result, their personal characteristics play a significant role in influencing decision-making, which ultimately impacts corporate behavior. Among various managerial attributes, the cognitive ability of entrepreneurs emerges as a crucial factor affecting strategic decisions. Stronger cognitive abilities enable entrepreneurs to better grasp the transformative potential of digital technology and make more informed decisions regarding its implementation, thereby enhancing its effectiveness in alleviating corporate financing constraints.

Furthermore, an entrepreneur's cognitive ability also has an impact on the firm's innovation activities. Entrepreneurs with high cognitive abilities are more open to new ideas and demonstrate a stronger inclination towards innovation and change (Wiersema and Bantel, 1992). This characteristic directly contributes to enterprise innovation (Camelo et al., 2010). For example, Wally and Baum (1994) utilized CEOs' educational levels as a proxy for cognitive ability and found that CEOs

with higher cognitive abilities were better equipped to analyze and process information, leading to more effective innovation.

Given the significant impact of entrepreneurial cognitive ability on technology adoption and innovation, this paper further investigates the heterogeneity effect of varying levels of cognitive ability on financing constraints. To operationalize this concept, we use the educational background of the entrepreneur as a proxy for cognitive ability (*Edu*). Specifically, if the legal representative of the enterprise holds a graduate degree or higher, their cognitive ability is considered strong and is coded as 1; otherwise, it is coded as 0. This dummy variable is then interacted with the original explanatory variable. The empirical results are presented in Table 5 (1). The results indicate that the coefficient of the interaction term is significantly negative at the 1 % level, suggesting that the stronger the entrepreneur's cognitive ability, the greater the mitigating effect of digital technology on financing constraints. This finding underscores the pivotal role of entrepreneurial cognitive ability in leveraging new technologies effectively.

4.3.2. Industry competition intensity

SMEs often face intense market competition in their early stages of development, which can significantly impact their decision-making and information disclosure practices. The level of competition within the industry can influence how companies disclose information, as competitors may mimic this disclosure and potentially harm the disclosing entity (Verrecchia, 1990). Moreover, SMEs often struggle to secure internal financing due to substantial funding gaps amidst high industry competition. Bhattacharya et al. (2003) observed that higher levels of information transparency lead to lower financing costs. Therefore, in highly competitive industries, firms have a strong incentive to proactively disclose more information to mitigate internal and external information asymmetry. Darrough and Stoughton (1990) argue that information disclosure can facilitate financing, suggesting that firms in fiercely competitive product markets may engage in higher quality information disclosure practices.

The application of digital technology can help reduce both internal and external information asymmetry, particularly in highly competitive industries. Furthermore, firms operating in such competitive markets are constantly at risk of market exit, driving them to pursue innovative activities. These innovations serve as high-quality signals to the market by showcasing positive growth prospects and potentially securing financial support at lower costs.

Based on the aforementioned analysis, this paper conducts further investigation into the heterogeneity effect of industry competition intensity on corporate financing. The Herfindahl Index (*HHI*), which is derived from the total assets of enterprises, is utilized as a proxy variable for industry competition intensity. This index gauges the level of competition within the industry for a specific year, with lower values indicating higher competition intensity. Subsequently, this variable is interacted with the original explanatory variable, and eq. (1) is employed for regression analysis. The empirical findings are presented in Table 5 (2). It can be observed that the coefficient of the interaction term is significantly positive, signifying that as the Herfindahl Index increases (and competition decreases), the impact of digital technology in alleviating financing constraints diminishes. This outcome suggests that in highly competitive industries, the application of digital technology can substantially enhance its effectiveness in mitigating financing constraints.

Table 5
Heterogeneity analysis.

	(1)	(2)	(3)
	FC	*FC*	*FC*
Digital	−0.0559***	−0.0784***	−0.0541***
	(0.0052)	(0.0047)	(0.0037)
Digital×Edu	−0.0232***		
	(0.0075)		
Edu	−0.0452***		
	(0.0039)		
Digital×Compete		0.2181***	
		(0.0746)	
Compete		0.0633	
		(0.0557)	
Digital×Scale			−0.1115***
			(0.0231)
Scale			−0.4678***
			(0.0162)
Control variables	Yes	Yes	Yes
Industry Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Within R^2	0.2380	0.2344	0.2954
N	49,600	49,600	49,600

4.3.3. Scale characteristics

SMEs form the backbone of private enterprises and are a vital driver of economic development. However, in practice, SMEs often face credit discrimination and struggle with financing constraints, which are not effectively alleviated despite the advantages large enterprises enjoy in accessing finance (Erdogan, 2019). With the advancement of the digital economy and the widespread adoption of digital technology, it remains to be proven whether SMEs can leverage digital technology to mitigate credit discrimination and alleviate their financing constraints. Large enterprises typically benefit from more standardized financial systems and greater external stakeholder oversight, resulting in higher information quality compared to SMEs. Nevertheless, the application of digital technology offers SMEs significant opportunities to enhance their own information transparency and transmit high-quality signals.

To investigate this issue, this paper first categorizes enterprises based on the Classification Standards for Large, Medium, Small, and Micro-sized Enterprises in Statistical Practices (2017) issued by the National Bureau of Statistics of China. We then introduce an enterprise size dummy variable (*Scale*), which is marked as 1 if the enterprise is classified as large and 0 otherwise. The empirical results are presented in Table 5 (3). The estimated coefficient of the interaction term between the enterprise size dummy variable and the application of digital technology is significantly negative, indicating that the application of digital technology by large enterprises can more effectively alleviate their financing constraints. Meanwhile, the original explanatory variable remains significantly negative at the 1 % level, demonstrating that the application of digital technology by micro, small, and medium-sized enterprises can also alleviate their financing constraints. These findings highlight the potential for SMEs to expand their financing opportunities through the strategic use of digital technologies.

4.4. Robustness tests

Although the suppressive effect of firms' application of digital technology on their financing constraints has been largely confirmed through the basic regressions presented in the previous section, there may be unobservable factors that influence these results and, consequently, affect the robustness of the findings. Therefore, it is necessary to conduct robustness tests to ensure the reliability of the initial conclusions.

4.4.1. Substitute the dependent variable

Although the existing literature provides a variety of methods to measure financing constraints, the lack of detailed financial data, such as corporate cash flow, prevents this study from constructing more complex comprehensive indicators like the WW index and KZ index. However, the ZHSP dataset includes other relevant indicators that can serve as alternative proxies for measuring corporate financing constraints. Corporate external financing can be broadly categorized into bond financing and equity financing. The dataset contains relevant indicators for external equity financing obtained by enterprises.

Some studies have demonstrated that enterprise innovation, as an

observable high-quality signal, is more likely to attract external venture capital (Mann and Sager, 2007; Hoenen et al., 2014). Therefore, this paper employs the amount of venture investment (*Venture*) and the amount of new equity financing in the current year (*Equity*) as proxy indicators to test the relationship. The empirical results are shown in columns (1) and (2) of Table 6. It can be observed that the coefficient of digital technology application is significantly positive at the 1 % level, indicating that the application of digital technology enhances the probability and opportunity for enterprises to secure external investment, thereby promoting corporate financing.

4.4.2. Propensity score matching

To address endogeneity issues, such as sample selection bias, this paper employs propensity score matching (PSM) combined with a staggered DID approach to reduce estimation bias and other problems arising from endogeneity. Control variables are utilized as covariates for propensity score matching, and the original sample is re-matched using a 1:6 nearest neighbor matching method within a 0.01 caliper. The estimated average treatment effect of applying digital technology was 3.2230 and significant at the 1 % level. The standardized deviations after matching are clustered around zero, indicating that the matched samples do not differ significantly in terms of matching variables, and the matching results are well balanced.

Based on the matched weights and matched samples, the regression of model (1) was re-run. The results are presented in column (3) of Table 6. The coefficient of *PSM_Digital* remains significantly negative at the 1 % significance level, indicating that the application of digital technology by enterprises has a significant inhibitory effect on their financing constraints. This further confirms the robustness of the findings in this paper.

4.4.3. Instrumental variables

Given that firm size, age, and industry are critical factors influencing firms' adoption of digital technology, this study utilizes the proportion of firms applying digital technology within the same scale-age-industry category as an instrumental variable. The rationale for this choice is as follows: The proportion of firms applying digital technology within the same scale, age, and industry is expected to be correlated with the likelihood of a target firm adopting digital technology. This correlation arises from firm-specific characteristics such as size and age, as well as industry-specific characteristics. Existing literature supports the presence of a peer-group effect at the firm level (Kaustia and Rantala, 2015). As technological innovation is crucial for a firm's future survival and development, firms tend to observe and respond similarly to the technological innovation strategies of peer-group firms (Matray, 2021). Therefore, it is believed that this peer-group effect influences firms' adoption of digital technology, which manifests as the correlation of this instrumental variable. Additionally, the degree of digital technology adoption by firms in a specific industry reflects demand characteristics for digital technology at the industry level—considered exogenous relative to individual firms. To further enhance exogeneity, we lagged the instrumental variable by one period.

Based on the analysis, this paper employs two-stage least squares (2SLS) regression for testing. The Kleibergen-Paap rk LM statistic is 1119.257, significantly passing the test at the 1 % level. The Cragg-Donald Wald F-statistic is 2003.38, and the critical value for the Stock-Yogo test at the 10 % level is 16.38, significantly rejecting the null hypothesis that the instrumental variable is a weak instrument. The results of the first and second stages are shown in columns (4) and (5) of Table 6. The coefficient in the second-stage regression is significantly negative, indicating that the application of digital technology by firms significantly alleviates their financing constraints. This conclusion is consistent with the original regression results, demonstrating the robustness of the research findings.

4.4.4. Parallel trend test

The key premise for the DID model to provide credible estimates of policy effects is that the parallel trend assumption holds true prior to the intervention or shock. To verify this assumption, the following model is set up for testing:

$$FC_{it} = \beta_0 + \sum_{K=-5, K\neq -1}^{5} \delta_K D_{i,T_{0+K}} + \lambda_i + \theta_t + \varepsilon_{it} \quad (3)$$

Where D is a dummy variable indicating whether digital technology is applied, with $D = 1$ for firms in the experimental group and $D = 0$ for firms in the control group. T_0 denotes the year in which the firm begins applying digital technology, and K represents the K-th year after the application of digital technology. δ_K denotes whether there is a significant difference in financing constraints between the experimental and control groups in the K-th year after the firm implements digital

Table 6
Robustness test.

	(1)	(2)	(3)	(4)	(5)
	Venture	*Equity*	*FC*	*Digital*	*FC*
Digital	0.0917***	0.1250***			−0.9142***
	(0.0138)	(0.0146)			(0.0326)
PSM_Digital			−0.0703***		
			(0.0040)		
IV				0.8474***	
				(0.0175)	
Income	0.0018	0.0287***	−0.0897***	0.0105***	−0.0860***
	(0.0024)	(0.0028)	(0.0012)	(0.0012)	(0.0019)
Cap	−0.0136***	0.0149***	−0.0262***	−0.0075***	−0.0461***
	(0.0025)	(0.0032)	(0.0012)	(0.0014)	(0.0019)
Profit	−0.3463***	−0.1848***	0.1516***	−0.0722***	0.0458***
	(0.0257)	(0.0171)	(0.0059)	(0.0093)	(0.0123)
Roe	−0.0221***	−0.0104**	−0.0029	−0.0053*	−0.0072*
	(0.0072)	(0.0047)	(0.0021)	(0.0032)	(0.0039)
Lev	−0.1593***	−0.1044***	−0.0101***	−0.0210***	−0.0361***
	(0.0099)	(0.0077)	(0.0033)	(0.0045)	(0.0057)
R&D_Staff	0.1001***	0.0747***	−0.0397***	0.1688***	0.1385***
	(0.0131)	(0.0130)	(0.0054)	(0.0079)	(0.0113)
SOE	−0.0529***	−0.0725***	−0.0555***	−0.0141*	−0.0685***
	(0.0103)	(0.0198)	(0.0085)	(0.0074)	(0.0113)
Industry Fixed Effect	YES	YES	YES	YES	YES
Time fixed effect	YES	YES	YES	YES	YES
N	49,515	49,515	45,726	33,514	33,514

technology. The meanings of other variables remain the same as previously described.

The sample period spans a total of 10 periods, including the five years preceding and the five years following the application of digital technology. To avoid the influence of multicollinearity, this study takes the first year before the application of digital technology as the base period. Model (3) is then used for regression analysis.

The results of the parallel trend test are shown in Table 7. It can be observed that the coefficients are not statistically significant in the period before the application of digital technology by firms, but become significantly negative after the application of digital technology. This indicates that there is no significant difference between the experimental group and the control group before the firm applies digital technology, confirming the satisfaction of the parallel trend hypothesis. However, after the application of digital technology, the financing constraints of the experimental group are significantly lower, which suggests that the application of digital technology has a significant mitigating effect on financing constraints. This finding reinforces the robustness of the original conclusion.

4.4.5. Placebo test

In order to further verify the meaningfulness of the regression results and to exclude potential interference from other factors, a placebo test was conducted. Specifically, the procedure involved the following steps: (1) Samples of the same proportion were randomly selected according to the proportion of samples with actual applications of digital technology and designated as the pseudo-experimental group. (2) The year in which the company implemented digital technology was randomly assigned. (3) Model (1) was re-applied for regression to obtain the pseudo-regression coefficient.

This process was repeated 500 times to exclude the interference of chance bias and to generate a distribution of pseudo-regression coefficients for the placebo test. The results are illustrated in Fig. 1, showing that the pseudo-regression coefficients follow a normal distribution centered around 0. The values of the pseudo-regression coefficients range approximately between −0.025 and 0.025. In contrast, the true regression coefficient from column (2) in Table 3 is −0.0706, which lies far outside the distribution range of the placebo regression coefficients. This indicates that the randomly assigned application of digital technology and the randomly assigned year of implementation have no significant impact on the financing constraints of enterprises, thereby confirming the robustness of the empirical results.

Table 7
Parallel Trend Test.

Periods	FC
T_{-5}	0.0206
	(0.0247)
T_{-4}	0.0040
	(0.0169)
T_{-3}	−0.0003
	(0.0136)
T_{-2}	−0.0110
	(0.0100)
T_0	−0.1014***
	(0.0057)
T_{+1}	−0.0763***
	(0.0067)
T_{+2}	−0.0452***
	(0.0080)
T_{+3}	−0.0214**
	(0.0105)
T_{+4}	−0.0333**
	(0.0139)
T_{+5}	−0.0705***
	(0.0232)
Control variables	Yes
Industry Fixed Effect	YES
Time fixed effect	YES
Within R^2	0.2354
N	49,600

5. Analysis of impact mechanisms

5.1. Mediation models

Considering financing constraints as the dependent variable, digital technology adoption as the independent variable, and both firm innovation capability and integration into innovation networks as mediating variables, this section proceeds to conduct an empirical investigation. The model employed is delineated as follows:

$$M_{it} = a_0 + a_1 Digital_{it} + a_2 X_{it} + \lambda_{1i} + \theta_{1t} + \varepsilon_{1it} \quad (4)$$

$$FC_{it} = c_0 + c_1 Digital_{it} + c_2 Digital_{it} \times M_{it} + c_3 M_{it} + c_4 X_{it} + \lambda_{2i} + \theta_{2t} + \varepsilon_{2it} \quad (5)$$

Where M_{it} signifies the mediating variable, and the remaining variables retain their meanings as delineated in Eq. (1). Initially, we employ Eq. (4) to ascertain the effect of firms' adoption of digital technologies on their innovation capabilities and integration into innovation networks. Should the estimated coefficient a_1 in Eq. (4) exhibit a significant positive value, this would indicate that the implementation of digital technologies by firms can notably enhance their innovation capabilities and facilitate their integration into innovation networks.

Thereafter, we interact the mediating variable M_{it} with the explanatory variable and then use Eq. (5) for estimation. If the coefficient c_2 of the interaction term is significantly negative, it implies that the stronger the enterprise's innovation ability and the deeper its participation in the innovation network, the more effectively digital technology adoption can alleviate its financing constraints. A significantly positive a_1 coupled with a significantly negative c_2 indicates that the adoption of digital technology can bolster a company's innovation capabilities and facilitate its integration into innovation networks, effectively signaling high quality to external parties, attracting external investment, and alleviating financing constraints.

Regarding the setting of the mechanism variable M_{it} for enterprise innovation capability, we initially measure it in terms of enterprise innovation performance and innovation achievement transformation. In this paper, the number of invention patent applications (*Patent*) and the number of new product sales (*Product*) are utilized as measures for the aforementioned mechanism.

Innovation efficiency is also an important indicator to measure the innovation ability of enterprises. Enterprise innovation efficiency is a direct expression of the relationship between R&D input and R&D output. Higher innovation efficiency indicates a firm's strong ability to integrate and utilize internal resources and higher marginal output of innovation. This can serve as a positive signal to the outside world about the firm's innovation capability and management efficiency, attracting external investors to expand their investment and enhancing the availability of corporate financing. Furthermore, higher innovation efficiency will also increase the enthusiasm of enterprises for R&D, maintain their technological and competitive advantages in the market, bring them rich profit returns, gain investors' favor and attention, and further promote their R&D innovation, forming a virtuous circle. Therefore, this paper also examines the effect of innovation capability from the perspective of innovation efficiency (*Efficiency*), which is expressed as the logarithm of the ratio of the sum of new product sales and technology contract turnover to R&D investment.

The scope of innovation networks is relatively broad, encompassing not only innovation cooperation between enterprises and other enterprises, universities, and research institutions, but also the participation of enterprises in industrial associations and industrial alliances. In terms of index measurement, this paper first uses the total expenditure of R&D funds entrusted by enterprises to other enterprises, universities, and research institutions as the index of enterprise innovation cooperation

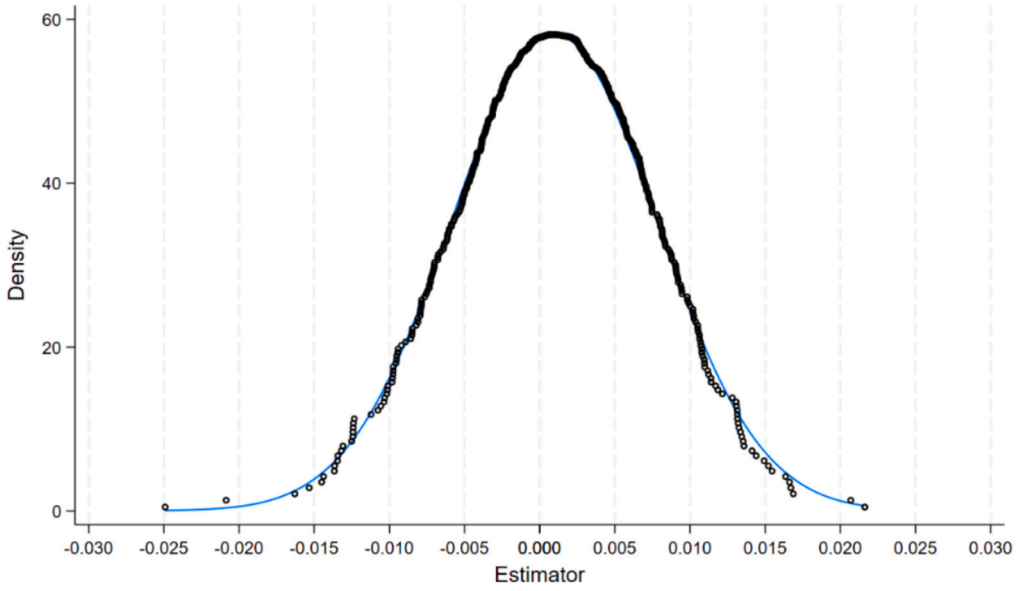


Fig. 1. Placebo test.

(*Networks*). Secondly, this paper also constructs a dummy variable (*Alliances*); if the enterprise joins an industrial alliance or industrial association, the value of the dummy variable is recorded as 1, otherwise, it is 0. These two indicators are then interacted with the original explanatory variables and tested again.

5.2. Empirical results and analysis

The empirical results of the mediating role of enterprise innovation capability are shown in Table 8. As can be seen from Panel A, the estimated coefficients of digital technology application on the three mechanism variables—number of invention patent applications, new product sales revenue, and innovation efficiency—are all significantly positive. This indicates that the application of digital technology can improve the innovation performance of enterprises, promote the transformation of innovation results, and enhance the innovation efficiency of enterprises, thereby providing a strong incentive for the innovation development of enterprises.

According to the regression results in Panel B, the coefficients of each interaction term are significantly negative, indicating that the innovation capability of enterprises can serve as an effective signal of the technical quality and capability of enterprises. The application of digital technology strengthens this signal effect, thus helping enterprises to obtain more financing opportunities and reduce financing constraints.

The empirical results related to innovation networks are shown in Table 9. In Table 9, the application coefficient of digital technology in columns (1) and (3) is significantly positive, indicating that the application of digital technology can promote the establishment of cooperative relations between enterprises and relevant external institutions, and help enterprises join industrial alliances and industrial associations, thereby facilitating their integration into the innovation network.

From the results in columns (2) and (4), it can be seen that the coefficients of the interaction terms are significantly negative, indicating that the integration of enterprises into the innovation network can be effectively transmitted as a signal of the technical quality of enterprises, thus helping to alleviate the financing constraints of enterprises. Therefore, the innovation network, as another embodiment of enterprise innovation ability, can also serve as a strong quality signal to help enterprises alleviate their own financing constraints.

Table 8
Test results of mediating effect of enterprise innovation capability.

Panel A. The effect of digital technology application on enterprise innovation capability

	(1)	(2)	(3)
	Patent	Product	Efficiency
Digital	0.1242***	0.3001***	0.0627***
	(0.0086)	(0.0329)	(0.0087)
Control variables	Yes	Yes	Yes
Industry Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Within R^2	0.1207	0.0710	0.0451
N	49,600	49,600	49,600

Panel B. The effect of firm innovation capability on financing constraint

	(1)	(2)	(3)
	FC	*FC*	*FC*
Digital	−0.0496***	−0.0515***	−0.0626***
	(0.0040)	(0.0042)	(0.0043)
Patent×Digital	−0.0144**		
	(0.0059)		
Patent	−0.1256***		
	(0.0044)		
Product×Digital		−0.0134***	
		(0.0014)	
Product		0.0025***	
		(0.0007)	
Efficiency×Digital			−0.0250***
			(0.0052)
Efficiency			0.0187***
			(0.0025)
Control variables	Yes	Yes	Yes
Industry Fixed Effect	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes
Within R^2	0.2838	0.2368	0.2353
N	49,600	49,600	49,600

6. Conclusion and discussion

6.1. Research conclusions

The application of digital technology can significantly ease the financing constraints of enterprises. Based on the enterprise dataset from

Table 9
Test results of mediating effect of innovation network.

	(1)	(2)	(3)	(4)
	Networks	FC	Alliances	FC
Digital	0.3377***	−0.0627***	0.0690***	−0.0612***
	(0.0266)	(0.0042)	(0.0036)	(0.0043)
Networks×*Digital*		−0.0036*		
		(0.0019)		
Networks		−0.0113***		
		(0.0012)		
Alliances×*Digital*				−0.0473***
				(0.0108)
Alliances				0.0026
				(0.0059)
Control variables	Yes	Yes	Yes	Yes
Industry Fixed Effect	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes
Within R^2	0.0624	0.2387	0.0706	0.2346
N	49,600	49,600	49,600	49,600

ZHSP, this paper empirically tests the impact of enterprise digital technology application on financing constraints using a staggered DID model. The findings reveal that the application of digital technology can effectively alleviate the financing constraints of enterprises. Moreover, the application of different types of digital technology and varying levels of digital technology adoption have differential effects on financing constraints.

Innovation ability and integration into innovation networks are crucial pathways through which digital technologies alleviate financing constraints. Given that corporate innovation serves as a high-quality signal that can be transmitted to external stakeholders, this paper examines the impact mechanism of digital technology on financing constraints from two perspectives: corporate innovation ability and integration into external innovation networks. The results indicate that the application of digital technology enhances the innovation ability of enterprises and promotes their integration into innovation networks, attracting external investment by effectively transmitting high-quality signals to the market, thereby alleviating their financing constraints.

The differences in enterprise characteristics and the development environment lead to varied heterogeneous effects of digital technology application on enterprise financing constraints. In this paper, the heterogeneous effects of entrepreneurial cognitive ability, industry competition, and firm size are discussed. The findings suggest that when entrepreneurial cognitive ability is stronger, industry competition is higher, and the firm size is larger, the application of digital technology has a stronger mitigating effect on financing constraints.

6.2. Theoretical contributions

The theoretical contribution of this paper is mainly concentrated in three aspects. First, consistent with existing literature (Xue et al., 2022; He et al., 2023; Guo et al., 2023), this paper posits that enterprises applying digital technology can reduce information asymmetry between themselves and external stakeholders, thereby aiding in obtaining financing. Unlike previous studies, this paper focuses specifically on alleviating financing constraints for SMEs and utilizes sample data from non-listed companies, which more broadly represents the impact of digital technology adoption among SMEs. Additionally, the indicators of digital technology usage in this study are derived from direct surveys and statistics from relevant statistical departments, rather than word frequency statistics and text analysis of annual reports. This approach provides a clearer indication of whether an enterprise has applied digital technology, making a significant addition to the existing literature.

Secondly, drawing on signal theory, this paper explores the signaling role of innovation capability and innovation networks in the process of digital technology alleviating financing constraints. This is particularly relevant for SMEs, as reducing information asymmetry alone may not necessarily help them secure financing. They also need to convey high-quality signals to the external world, demonstrating their development potential and thus increasing the likelihood of investment. Patents and the integration of enterprises into innovation networks can serve as important signals for external investors (Stuart, 2000; Mann and Sager, 2007; Haeussler et al., 2012; Hsu and Ziedonis, 2013; Hoenen et al., 2014). Therefore, this paper examines SMEs financing from the perspective of enterprise innovation. The results confirm that enterprise innovation can indeed serve as a quality signal, aiding in securing financing.

Finally, building upon signal theory, this paper investigates heterogeneity from the perspectives of industry competition, entrepreneurial cognitive ability, and firm size. More intense industry competition can drive more active innovation activities and a stronger motivation to convey high-quality signals to the external world. Entrepreneurs with higher cognitive abilities are more likely to focus on the application of new technologies and the signal effects of innovation. Furthermore, some literature suggests that larger enterprises have more advantages in obtaining financing and alleviating financing constraints (Erdogan, 2019). Our research findings support this view regarding the size of enterprises. However, empirical results also demonstrate that digital technology can assist SMEs in alleviating financing constraints, providing a new pathway for SMEs to manage their financing challenges. These findings contribute significantly to the existing literature on the use of digital technologies and financing.

6.3. Managerial and policy implications

With the rise of the digital economy and against the backdrop of new opportunities and missions for digital transformation, this paper explores the primary pathways through which digital technology impacts enterprise financing constraints. This research provides valuable references for future corporate financing efforts and offers empirical evidence at the micro level for government departments to improve enterprise credit policies. Summarizing the above research findings, this paper proposes the following policy recommendations.

For SMEs, first, they should fully grasp the opportunities of digital transformation, leveraging the role of digital technology to empower business development. This includes promoting the large-scale integrated application of Big Data, Cloud Computing, IoT, AI, and other technologies, while focusing on precise digitalization and understanding the issues of technology selection and timing for their own digital transformation. Secondly, they should smooth the flow channels of data elements, taking the initiative to transmit reliable enterprise information to the outside world. This maximizes the advantages of digital technology in information transmission, reducing information barriers with external parties, and thus easing financing constraints. Lastly, they should emphasize the signaling role of innovation activities in corporate financing, using digital technology to effectively enhance their innovation capabilities, expand their own strength, and actively integrate into the innovation network. This helps in dispersing the risk of R&D and actively conveying their high-quality signals to the outside world to help obtain more financing opportunities.

For the government, necessary policy guidance and support are important avenues for promoting SMEs' financing. On one hand, the government needs to emphasize the synergistic effects brought by the application of digital technologies to SMEs, promoting deep integration, coexistence, and innovation between digital technologies and the real economy, with a focus on assisting SMEs that lack resources for digital transformation. On the other hand, the government should increase support for SMEs' digital transformation and R&D innovation through measures such as establishing innovation funds, R&D incentives, and tax benefits, promoting the organic integration of digital technology application and innovation in SMEs. Additionally, the government should enhance legal protection for SMEs during the innovation process by improving the intellectual property rights (IPR) protection system.

Strengthening the protection of patents related to digital technologies and software copyrights can reduce infringement occurrences, thereby effectively boosting SMEs' innovation motivation.

Despite the fact that government policy support can provide certain guarantees and incentives for SMEs' application of digital technologies and innovation activities, challenges and other potential impacts still exist in reality. First, while policies like innovation funds and tax benefits can support SMEs, ensuring precise allocation of funds and preventing misuse remains a critical issue in policy implementation. Moreover, since innovation is a long-term dynamic process, short-term policy support may not yield immediate results, so the government must also consider establishing dynamic and long-term policy support systems.

Furthermore, in terms of IPR protection, the development of digital technologies, especially artificial intelligence, may generate content similar to or derived from original works without authorization, blurring copyright boundaries and complicating infringement determinations. The formulation and release of IPR-related laws and regulations tend to lag behind, which may pose significant governance challenges during the development stage of digital technologies. Therefore, accelerating the legislative process, establishing flexible policy adjustment mechanisms, and encouraging industry self-regulation are important issues for government governance.

6.4. Limitations and future research

Limited by data and space, this study has some limitations, which will provide directions and ideas for future research. First, information asymmetry and principal-agent problems are the key factors that affect corporate financing constraints. While most of the existing literature uses data from listed companies to validate these issues, there is a lack of evidence on the extent to which digital technologies can mitigate information asymmetry in SMEs. Due to the limitation of data, we cannot construct variables of information asymmetry and principal-agent problems and conduct relevant empirical tests. Therefore, we can only theoretically infer that digital technology can alleviate the financing constraints of enterprises by reducing the degree of information asymmetry between SMEs and external stakeholders. Second, digital transformation is a process, not a static outcome. Similarly, the lack of comprehensive data prevents a detailed examination of the specific phases of digital transformation that are required to significantly impact financing constraints. From the limited data available, we can tentatively conclude that the more companies adopt digital technology, the more it helps ease their financing constraints. Third, this paper explores how digital technologies enhance enterprise innovation to promote corporate financing, focusing more on the synergistic effects of enterprises applying digital technologies. But in reality, different digital technologies have significant distinctions in promoting enterprise innovation and alleviating financing constraints. Which digital technology offers greater empowerment for enterprises in different industries and fields? Which combinations of digital technologies can achieve superior effects? What digital technologies should enterprises prioritize at different stages of their lifecycle? Due to length constraints, we did not delve further into these topics. We hope that the above three points provide directions and ideas for further research on the transformative impact of digital technologies on enterprises.

CRediT authorship contribution statement

Kang Wang: Resources, Funding acquisition, Conceptualization, Writing – original draft. **Kang Sun:** Writing – original draft, Visualization, Software, Data curation. **Yifei Li:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Qian Xie:** Writing – review & editing, Supervision, Project administration, Formal analysis.

Author statement

We declare that we have not submitted this manuscript to other publishers and there is no copyright dispute.

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Appendix A. Classification standards for large, medium, small, and micro-sized enterprises in statistical practices

Name of Industry	Indicator	Large-sized	Medium-sized	Medium-sized	Micro-sized
Agriculture	Total Operating Income(Y)	Y ≥ 20,000	500 ≤ Y<20,000	50 ≤ Y<500	Y<50
Industry	Employees of the Company(X)	X ≥ 1000	300 ≤ X<1000	20 ≤ X<300	X<20
	Total Operating Income(Y)	Y ≥ 40,000	2000 ≤ Y<40,000	300 ≤ Y<2000	Y<300
Construction Industry	Total Operating Income(Y)	Y ≥ 80,000	6000 ≤ Y<80,000	300 ≤ Y<6000	Y<300
	Total Assets(Z)	Z ≥ 80,000	5000 ≤ Z<80,000	300 ≤ Z<5000	Z<300
Wholesale Trade	Employees of the Company(X)	X ≥ 200	20 ≤ X<200	5 ≤ X<20	X<5
	Total Operating Income(Y)	Y ≥ 40,000	5000 ≤ Y<40,000	1000 ≤ Y<5000	Y<1000
Transportation Industry	Employees of the Company(X)	X ≥ 1000	300 ≤ X<1000	20 ≤ X<300	X<20
	Total Operating Income(Y)	Y ≥ 30,000	3000 ≤ Y<30,000	200 ≤ Y<3000	Y<200
Warehousing Industry	Employees of the Company(X)	X ≥ 200	100 ≤ X<200	20 ≤ X<100	X<20
	Total Operating Income(Y)	Y ≥ 30,000	1000 ≤ Y<30,000	100 ≤ Y<1000	Y<100
Postal Industry	Employees of the Company(X)	X ≥ 1000	300 ≤ X<1000	20 ≤ X<300	X<20
	Total Operating Income(Y)	Y ≥ 30,000	2000 ≤ Y<30,000	100 ≤ Y<2000	Y<100
Accommodation Industry	Employees of the Company(X)	X ≥ 300	100 ≤ X<300	10 ≤ X<100	X<10
	Total Operating Income(Y)	Y ≥ 10,000	2000 ≤ Y<10,000	100 ≤ Y<2000	Y<100
Catering Industry	Employees of the Company(X)	X ≥ 300	100 ≤ X<300	10 ≤ X<100	X<10
	Total Operating Income(Y)	Y ≥ 10,000	2000 ≤ Y<10,000	100 ≤ Y<2000	Y<100
Information Transmission Industry	Employees of the Company(X)	X ≥ 2000	100 ≤ X<2000	10 ≤ X<100	X<10
	Total Operating Income(Y)	Y ≥ 100,000	1000 ≤ Y<100,000	100 ≤ Y<1000	Y<100
Software and Information Technology Services	Employees of the Company(X)	X ≥ 300	100 ≤ X<300	10 ≤ X<100	X<10
	Total Operating Income(Y)	Y ≥ 10,000	1000 ≤ Y<10,000	50 ≤ Y<1000	Y<50
Real Estate Industry	Total Operating Income(Y)	Y ≥ 200,000	1000 ≤ Y<200,000	100 ≤ Y<1000	Y<100
	Total Assets(Z)	Z ≥ 10,000	5000 ≤ Z<10,000	2000 ≤ Z<5000	Z<2000
Property Management	Employees of the Company(X)	X ≥ 1000	300 ≤ X<1000	100 ≤ X<300	X<100
	Total Operating Income(Y)	Y ≥ 5000	1000 ≤ Y<5000	500 ≤ Y<1000	Y<500

(*continued on next page*)

(*continued*)

Name of Industry	Indicator	Large-sized	Medium-sized	Medium-sized	Micro-sized
Leasing and Business Services	Employees of the Company(X)	X ≥ 300	100 ≤ X<300	10 ≤ X<100	X<10
	Total Assets(Z)	Z ≥ 120,000	8000 ≤ Z<120,000	100 ≤ Z<8000	Z<100
Other Industries	Employees of the Company(X)	X ≥ 300	100 ≤ X<300	10 ≤ X<100	X<10

Notes

1.The unit of measurement for Total Operating Income and Total Assets is in ten thousand Yuan, while the unit of measurement for the Employees of the Company is in individuals.

2. Large-sized, medium-sized, and small-sized enterprises must simultaneously meet the lower limits of the listed indicators; otherwise, they will be downgraded by one level. Micro-sized enterprises only need to satisfy one of the listed indicators.

Data availability

The data that has been used is confidential.

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